10. Object List

- The objects are the second element that will be built by the radar and it is going to be the history of the clusters. By other meaning, the objects are the real subjects that we are interested in and can be defined as it is a punch of reflected signals. The nearest object will be built firstly and so on. With a rising number of objects the distance rises, too.
- The object list output consists of up to five message definitions that are sent in a regular interval if objects are selected in signal ***RadarCfg_OutputType*** (0x200).

1. ***Object_0_Status*** (0x60A) – The first message contains list header information, i.e. the number of objects that are sent afterwards
2. ***Object_1_General*** (0x60B) – This message contains the position and velocity of the objects and is sent repeatedly for all the tracked objects.
3. ***Object_2_Quality*** (0x60C) – This message contains the quality information of the objects and is only sent if it was activated in signal ***RadarCfg_SendQuality*** (0x200). It is sent repeatedly for all objects in the same way as message ***Object_1_General*** (0x60B).
4. ***Object_3_Extended*** (0x60D) – This message contains additional object properties and is only sent if it was activated in signal ***RadarCfg_SendExtInfo*** (0x200). It is sent repeatedly for all objects in the same way as message ***Object_1_General*** (0x60B).
5. ***Object_4_Warning*** (0x60E). This message contains the collision detection warning state and is only sent if collision detection was activated in message ***CollDetCfg*** (0x400). It is sent repeatedly for all objects in the same way as message ***Object_1_General*** (0x60B).

If the quality information, extended information and/or warning state is sent, first all messages of type ***Object_1_General*** (0x60B) are sent and afterwards all messages of type ***Object_2_Quality*** (0x60C), afterwards of type ***Object_3_Extended*** (0x60D) and/or afterwards of type ***Object_4_Warning*** (0x60E).

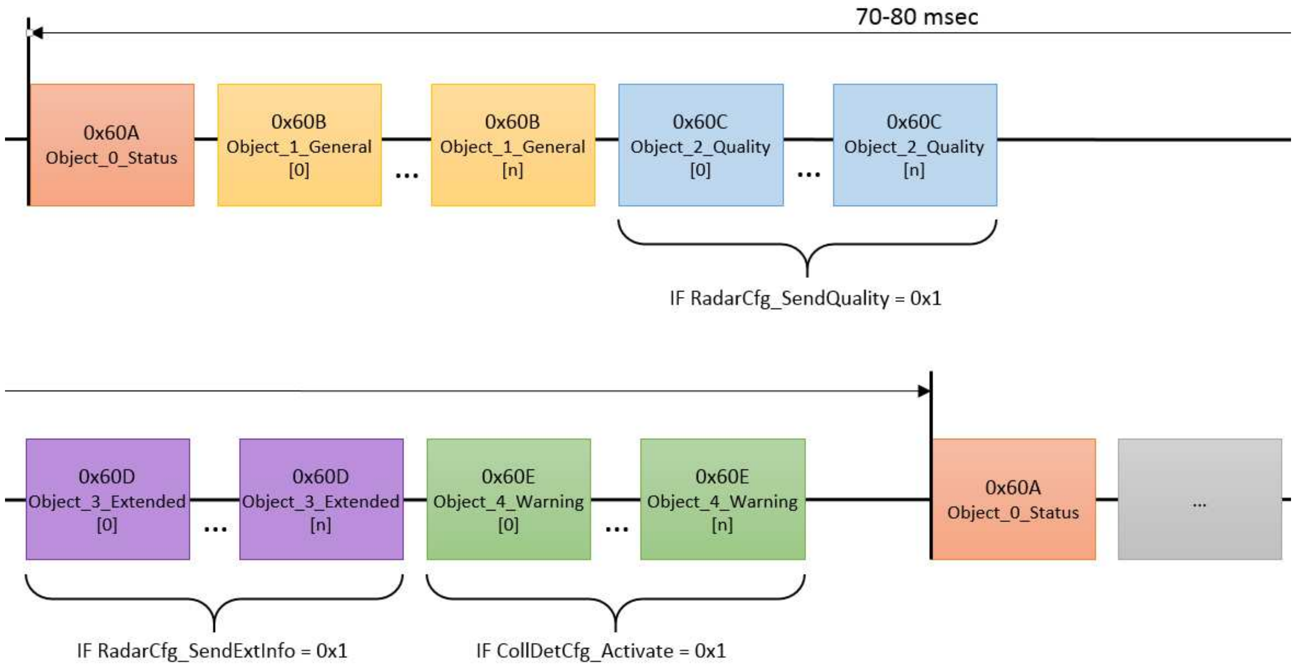

Figure 28: Overview of object list messages that are sent cyclically by the radar

10.1. Object list status (0x60A)

The message ***Object_0_Status*** (0x60A) contains the object list header information and is sent as first message of the cluster list output and only once per measurement cycle.

Figure 29: Object_0_Status - message layout (0x60A)

Table 37: Object_0_Status - message content (0x60A)

Signal	Start	Len	Min	Max	Res	Unit
Object_NofObjects	0	8	0	255	1	-
Object_MeasCounter	16	16	0	65535	1	-
Object_InterfaceVersion	28	4	0	15	1	always "1"

Table 38: Object_0_Status - signal description (0x60A)

Start	Signal	Description
0	Object_NofObjects	Number of objects (max. 100 Objects)
16	Object_MeasCounter	Measurement cycle counter (counting up since startup of sensor and restarting at 0 when > 65535)
28	Object_InterfaceVersion	Object list CAN interface Version-No. . It is always "**1**" till any Object Identifier will be changed in any coming SW-update.

10.2. Object general information (0x60B)

This message contains the position and velocity of the objects and is sent repeatedly for all the tracked objects.

Figure 30: Object_1_General - message layout (0x60B)

Table 39: Object_1_General - message content (0x60B)

Signal	Start	Len	Offset	Min	Max	Res	Unit
Object_ID	0	8		0	255	1	
Object_DistLong	19	13	-500	-500	+1138.2	0.2	m
Object_DistLat	24	11	-204.6	-204.6	+204.8	0.2	m
Object_VrelLong	46	10	-128.00	-128.00	127.75	0.25	m/s
Object_DynProp	48	3		0	7	1	0x0: moving 0x1: stationary 0x2: oncoming 0x3: stationary candidate 0x4: unknown 0x5: crossing stationary 0x6: crossing moving 0x7: stopped
Object_VrelLat	53	9	-64.00	-64.00	63.75	0.25	m/s
Object_RCS	56	8	-64.0	-64.0	63.5	0.5	dBm²

Table 40: Object_1_General - signal description (0x60B)

Start	Signal	Description
0	Object_ID	Object ID (since objects are tracked, the ID is kept throughout measurement cycles and does not have to be consecutive)
19	Object_DistLong	Longitudinal (x) coordinate
24	Object_DistLat	Lateral (y) coordinate
46	Object_VrelLong	Relative velocity in longitudinal direction (x) stationary candidates => vrel (y) made by own speed and yaw rate => rest of speed projected to (x) Math.: f_VrelY = - (f_DistX + LongPosToRot) * YawRate f_VrelX = (f_VrelRad - sin_(f_AzAngle) * f_VrelY)/cos_(f_AzAngle) otherwise (moving candidates): => projected to (x) Math.: f_VrelY = 0.0 f_VrelX = f_VrelRad/COS_(f_AzAngle) if cos_(f_AzAngle) = 0, => f_VrelRad is used
53	Object_VrelLat	Relative velocity in lateral direction (y) (see (x) above)
48	Object_DynProp	Dynamic property of the object indicating if the object is moving or stationary (this value can only be determined correctly if the speed and yaw rate is given correctly)
56	Object_RCS	Radar cross section

10.3. Object quality information (0x60C)

This message contains the quality information of the objects and is only sent if it was activated in signal ***RadarCfg_SendQuality*** (0x200). It is sent repeatedly for all objects in the same way as message ***Object_1_General*** (0x60B).

Figure 31: Object_2_Quality - message layout (0x60C)

Table 41: Object_2_Quality - message content (0x60C)

Signal	Start	Len	Min	Max	Res	Unit
Obj_ID	0	8	0	255	1	-
Obj_DistLong_rms	11	5	0	31	1	m
Obj_VrelLong_rms	17	5	0	31	1	m/s
Obj_DistLat_rms	22	5	0	31	1	m
Obj_VrelLat_rms	28	5	0	31	1	m/s
Obj_ArelLat_rms	34	5	0	31	1	m/s^2
Obj_ArelLong_rms	39	5	0	31	1	m/s^2
Obj_Orientation_rms	45	5	0	31	1	deg
Obj_MeasState	50	3	0	7	1	0x0: deleted 0x1: new 0x2: measured 0x3: predicted 0x4: deleted for merge 0x5: new from merge

Obj_ProbOfExist	53	3	0	7	1	0x0: invalid 0x1: <25% 0x2: <50% 0x3: <75% 0x4: <90% 0x5: <99% 0x6: <99.9% 0x7: <=100%

Table 42: Object_2_Quality – signal description (0x60C)

Start	Signal	Description
0	Obj_ID	Object ID (since objects are tracked, the ID is kept throughout measurement cycles and does not have to be consecutive)
11	Obj_DistLong_rms	Standard deviation of longitudinal distance
17	Obj_VrelLong_rms	Standard deviation of longitudinal relative velocity
22	Obj_DistLat_rms	Standard deviation of lateral distance
28	Obj_VrelLat_rms	Standard deviation of lateral relative velocity
34	Obj_ArelLat_rms	Standard deviation of lateral relative acceleration
39	Obj_ArelLong_rms	Standard deviation of longitudinal relative acceleration
45	Obj_Orientation_rms	Standard deviation of orientation angle (see: figure 3)
50	Obj_MeasState	Measurement state indicating if the object is valid and has been confirmed by clusters in the new measurement cycle: 0x0: Deleted Object – Object has been deleted – is displayed during the last cycles of transmission just before the object ID disappears. 0x1: New Object is created – is displayed during the first cycles of transmission just after the object ID is created. Can also be checked by the character *Object_MeasCounter*. 0x2: Measured – Object creation has been confirmed by the actual measurement. Cluster could be created. 0x3: Predicted - Object creation could not be confirmed by the actual measurement. Cluster could not be created. 0x4: Deleted for merge - Object became deleted in order to be merged with another Object. 0x5: New from merge – new Object became created after a merge.
53	Obj_ProbOfExist	Probability of existence

Table 43: Signal value table for Obj_Orientation_rms, Obj_DistLong_rms, Obj_DistLat_rms, Obj_VrelLong_rms, Obj_VrelLat_rms, Obj_ArelLat_rms, Obj_ArelLong_rms (0x60C).

Parameter	Values for signal Obj_Orientation_rms [deg]	Value for signals Obj_DistLong_rms, Obj_DistLat_rms [m] Obj_VrelLong_rms, Obj_VrelLat_rms [m/s] Obj_ArelLat_rms, Obj_ArelLong_rms [m/s²]
0x0	<0.005	<0.005
0x1	<0.007	<0.006
0x2	<0.010	<0.008
0x3	<0.014	<0.011
0x4	<0.020	<0.014
0x5	<0.029	<0.018
0x6	<0.041	<0.023
0x7	<0.058	<0.029
0x8	<0.082	<0.038
0x9	<0.116	<0.049
0xA	<0.165	<0.063
0xB	<0.234	<0.081
0xC	<0.332	<0.105
0xD	<0.471	<0.135
0xE	<0.669	<0.174
0xF	<0.949	<0.224
0x10	<1.346	<0.288
0x11	<1.909	<0.371
0x12	<2.709	<0.478
0x13	<3.843	<0.616
0x14	<5.451	<0.794
0x15	<7.734	<1.023
0x16	<10.971	<1.317
0x17	<15.565	<1.697
0x18	<22.081	<2.187
0x19	<31.325	<2.817
0x1A	<44.439	<3.630
0x1B	<63.044	<4.676
0x1C	<89.437	<6.025
0x1D	<126.881	<7.762
0x1E	<180.000	<10.000
0x1F	invalid	invalid

10.4. Object extended information (0x60D)

This message contains additional object properties and is only sent if it was activated in signal ***RadarCfg_SendExtInfo*** (0x200). It is sent repeatedly for all objects in the same way as message ***Object_1_General*** (0x60B).

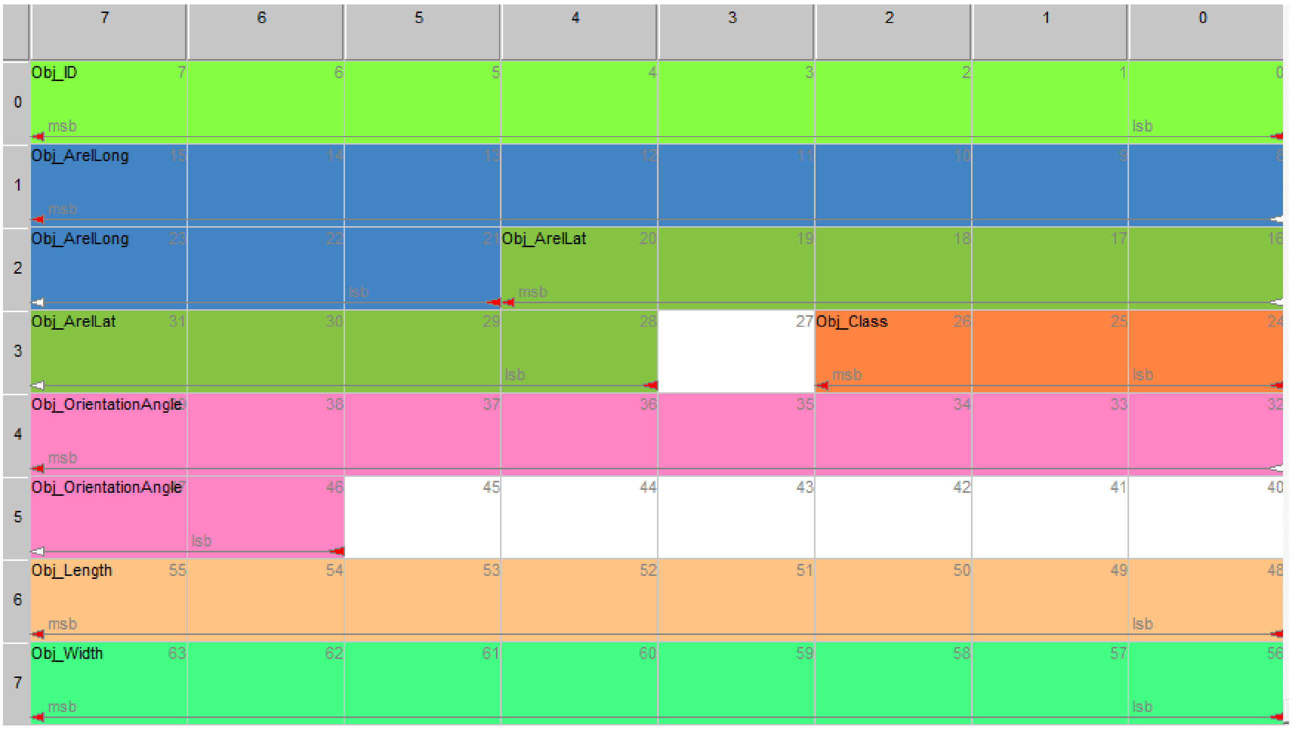

Figure 32: Object_3_Extended - message layout (0x60D)

Table 44: Object_3_Extended - message content (0x60D)

Signal	Start	Len	Offset	Min	Max	Res	Unit
Object_ID	0	8		0	255	1	
Object_ArelLong	21	11	-10.00	-10.00	10.47	0.01	m/s²
Object_Class	24	3		0	7	1	0x0: point 0x1: car 0x2: truck 0x3: not in use 0x4: motorcycle 0x5: bicycle 0x6: wide 0x7: reserved
Object_ArelLat	28	9	-2.50	-2.50	2.61	0.01	m/s² always "0"
Object_OrientationAngel	46	10	-180.00	-180.00	180.00	0.4	deg
Object_Length	48	8		0.0	51.0	0.2	m
Object_Width	56	8		0.0	51.0	0.2	m

Table 45: Object_3_Extended - signal description (0x60D)

Start	Signal	Description
0	Object_ID	Object ID (since objects are tracked, the ID is kept throughout measurement cycles and does not have to be consecutive)
21	Object_ArelLong	Relative acceleration in longitudinal direction
24	Object_Class	0x0: point 0x1: car 0x2: truck 0x3: not in use 0x4: motorcycle 0x5: bicycle 0x6: wide 0x7: reserved
28	Object_ArelLat	Relative acceleration in lateral direction (y) This value is permanently set to “Zero” see Cluster_VrelLat (Table 33)
46	Object_OrientationAngel	Orientation angle of the object (see: figure 3 - picture above); the change of an angle created by turning movement of a tracked obstacle over time. The creation of that value always starts at “0” and increases depending on a certain rotation of the obstacle. It keeps “0” if a rotation does not happen.
48	Object_Length	Length of the tracked object
56	Object_Width	Width of the tracked object

10.5. Object collision detection warning (0x60E)

This message contains the collision detection warning state and is only sent if collision detection was activated in message ***CollDetCfg*** (0x400). It is sent repeatedly for all objects in the same way as message ***Object_1_General*** (0x60B).

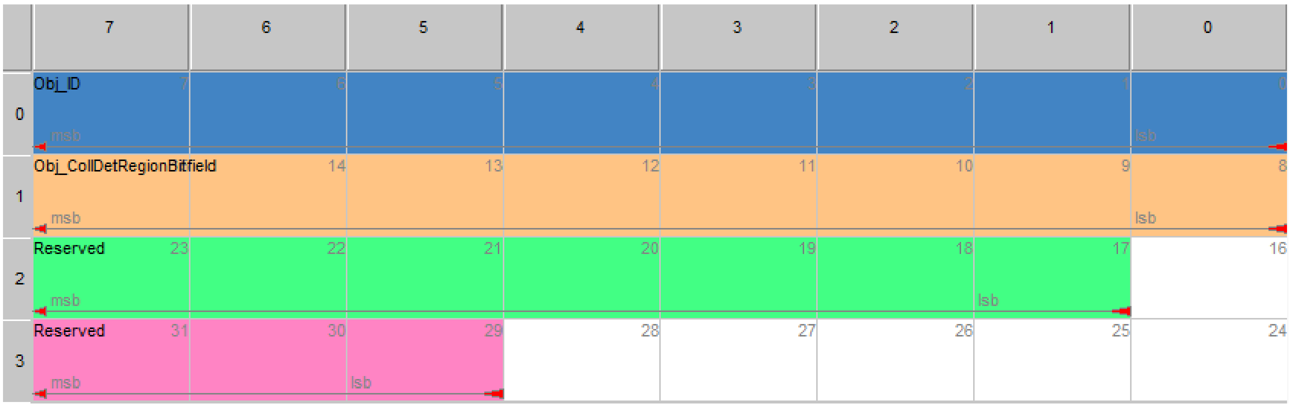

Figure 33: Object_4_Warning - message layout (0x60E)

Table 46: Object_4_Warning - message content (0x60E)

Signal	Start	Len	Min	Max	Res	Unit
Object_ID	0	8	0	255	1	
Object_CollDetRegionBitfield	8	8	0	255	1	

Table 47: Object_4_Warning - signal description (0x60E)

Start	Signal	Description
0	Object_ID	Object ID (since objects are tracked, the ID is kept throughout measurement cycles and does not have to be consecutive)
8	Object_CollDetRegionBitfield	Bit field of the regions, with bit set to 1 for regions that have a collision with this object